

Synth Benchmark Pipeline – end to end

Benchmarking the Synth harness (on Hermes) across DD · CFO · Tax · Legal, against frontier models, under determinism-by-design

START

business task definitions & client questionnaires

1 DATASET & QUESTIONNAIRE

Task / questionnaire
authoring

Golden benchmark (versioned CSV, 76 tasks)
agent.prompt · expected.behavior · expected.tool.calls
output.format · rubric · side.effect.checks
reference.answer · grading.signals

Offline fixtures
PDF · scans · CSV · .md
seeded into workspace

Connected / live data
OneDrive sync → harness
(Zoho / Tally / MS)

Task labels & routing keys: service line {DD · CFO · Tax · Legal} · BI band {Junior 0–3 · Senior 4–7} · complexity {Easy · Med · Hard} · gradability {Deterministic 50 · Hybrid 16 · Judgment 10}

2 PROVISIONING & DETERMINISM SETUP

Hermetic workspace
builder
seed intake/, COA, masters

Sandbox tenants
Zoho · Tally · OneDrive · MS Graph
snapshot baseline state S_0

Run manifest
pinned model IDs · Synth commit · tool versions
prompt templates · temp = 0 · seeds

Immutable input store
content-addressed

3 RUN MATRIX (cartesian product) & SCHEDULING

models →

Sonnet 4.6 / 4.7

Opus 4.8

GPT-5.5 Codex

Gemini 3.5 Pro

Hermes

[Keep + Fabric]

harness →

Synth – full

Synth – ablations

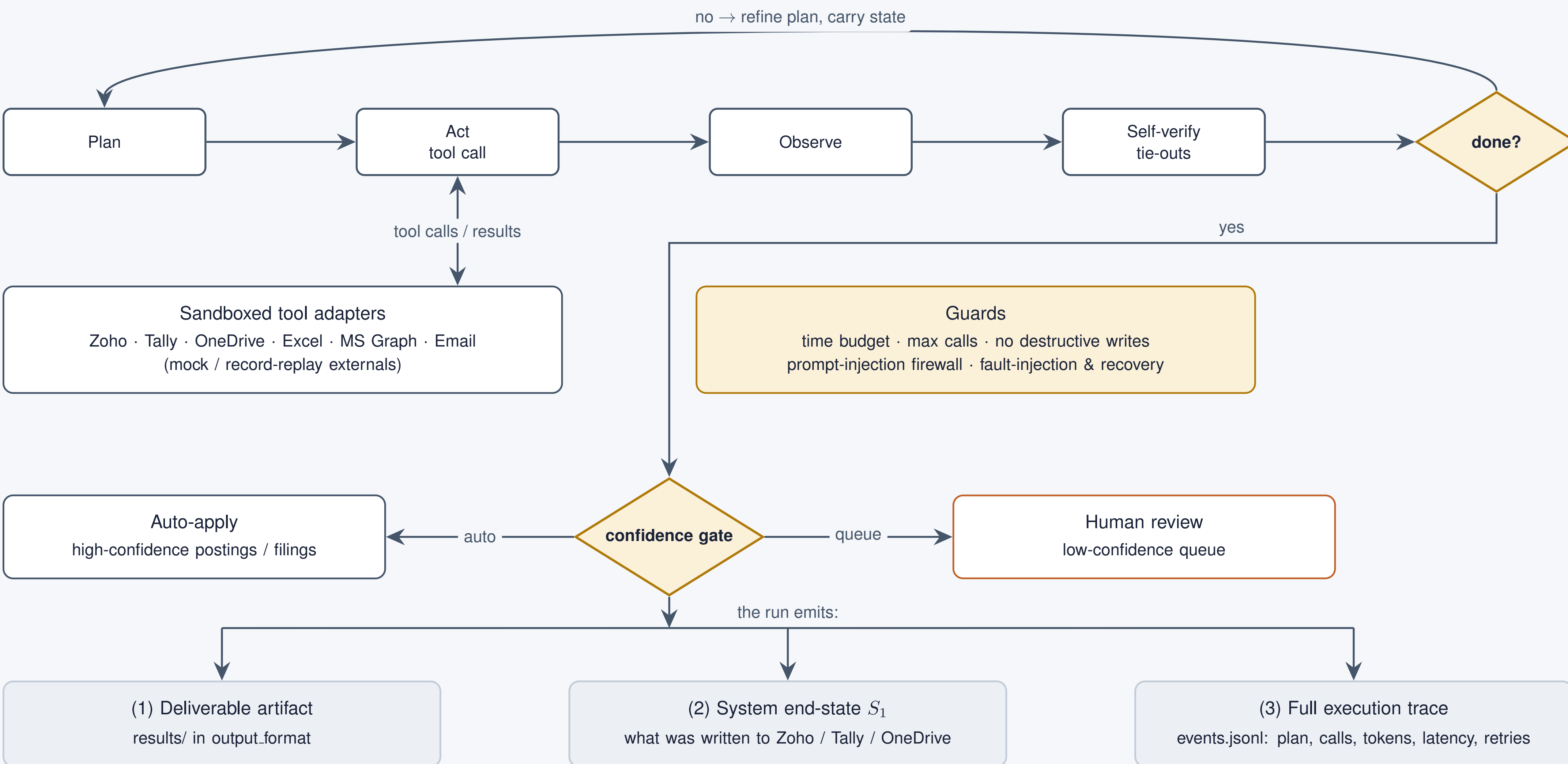
Baseline ReAct

grid cell = (model × harness) × 76 tasks × N seeds ⇒ bounded-variance sampling (run each cell N times)

Job scheduler / dispatcher – emits one isolated, replayable cell per run (forked workspace + tenant from S_0)

4 AGENT EXECUTION (Synth on Hermes, per cell)

Synth harness drives the loop below over the sandboxed tools, under the guards



5 CAPTURE & NORMALIZE

Collector + normalizer
parse output.format → canonical records

State diff engine
 S_1 vs S_0 vs expected side effects

Immutable run store
fully replayable: manifest + trace + outputs

6 GRADING (graders compare against golden references from stage 1)

grader router

Rule-based grader
artifact vs reference.answer:
exact · numeric $\pm$ tol ·
set.match · contradiction

State-audit grader
state diff vs side.effect.checks

LLM judge
rubric-scored; frozen model;
multi-sample agreement

Per-task graded result = pass / fail + per-signal sub-scores (rule, state, judge)

routing by gradability: **Deterministic** → Rule + State · **Hybrid** → Rule + State + Judge · **Judgment** → Judge + State

7 METRICS (≈19-metric scorecard)

Correctness & coverage
task success (pass%) · field accuracy ·
numeric tie-out · tool-call
F1 · findings recall / precision

Reliability & determinism
consistency across N seeds ·
schema adherence · self-verify rate ·
injection-resistance · destructive-rate = 0 · recovery

Efficiency & cost
latency vs time.budget.s · tool-call
efficiency · tokens / \$ per task ·
autonomy & intervention rate

Calibration & economics
confidence calibration · expert-time
leverage (expert.mins / agent.time) ·
BI-weighted value

Variance & confidence intervals computed across the N seeds of every cell (a “Deterministic” task that varies across seeds raises a flag)

8 BENCHMARK & DECISION

Aggregation engine
per (model × harness); slice by line / category /
complexity / band; weight by BI band & expert-time

Leaderboard
composite Synth Score per
(model × harness), with CIs

Regression vs prior
Synth versions
drift detection on the frozen benchmark

readiness gate

hard gates: state-audit pass $\geq$ thr $\wedge$ destructive = 0
 $\wedge$ weighted success on Senior-Hard $\geq$ thr

GO / NO-GO
per service line
Kayess · Eldaas

Export
Beamer deck · dashboards

END

billable-deployment decision + benchmark report

* DETERMINISM-BY-DESIGN SUBSTRATE (cross-cuts every stage above)

pinned model / harness / tool versions · frozen content-addressed fixtures · seed & temperature control · full-trace logging (events.jsonl)
replay engine · bounded-variance monitors · golden references · immutable audit trail

Legend:

data store

process

rule grader

state grader

LLM judge

gate

⇒ main flow

→ feedback